Name (Last, First):

Rissman, Joseph

NEU ID:

U09545917

Assignment 8

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 45 = 3 + 42 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:
3 points: Correct answer with work shown
2 points: Incorrect answer but attempt shows some understanding (work shown)
1 point: Incorrect answer but an attempt was made (work shown)
0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts] (link to your live google doc)

https://docs.google.com/document/d/1dzAgyy7UhyrmR9CyIVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing

I. Confidence Intervals for Proportions (7 x 3 = 21 pts)

1) A live action role-playing game (LARP) is a form of role-playing game where the participants physically portray their characters in a fictional setting, while interacting with each other in character. There are game rules to dictate the interactions between players and to determine the winner of the games.

A local gamemaster would like to determine how many people in the area would be interested in participating in his LARPing event. He sent out a survey to 1500 people in the area and asked if they would be interested in participating. 432 people said that they would be interested in participating in a LARPing event.

(a) State the parameter our confidence interval will estimate.

Answer:

p = the proportion of people in the area would be interested in participating in a LARPing event

(b) Identify each of the conditions that must be met to use this procedure, and explain how you know that each one has been satisfied.

Answer:

Random: it's not stated that the survey is sent to random people, but it's implied.
10%: it's reasonable to assume 1500 people is less than 10% of people in the area
Large counts: The number of successes (432) and failures (1068) are at least 10

(c) Find the appropriate critical value and the standard error of the sample proportion.

Answer:

For a 95% confidence interval, the critical value is 1.96.
SE = sqrt(0.288*0.712/1500) = 0.0117

(d) Give the 95% confidence interval.

Answer:

0.288 ± 0.0117
(0.2763, 0.2997)

(e) Interpret the confidence interval constructed in part (d) in the context of the problem.

Answer:

95% confident that the interval from 0.2763 to 0.2997 captures p = the proportion of people in the area would be interested in participating in a LARPing event

(f) This poll was conducted through email. Explain how undercoverage could lead to a biased estimate in this case, and speculate about the direction of the bias.

Answer:

People without an email address would not be included in the survey (this is likely almost no one these days), but also people who wouldn't open an email with a subject including "LARP" would be under-covered. This would bias the results to seem like more people are into LARP than there actually is.

2) The EPA wants to conduct a survey to determine the current approval rating on the government's handling of environmental action. From previous studies, they know that they need to poll 275 people to achieve their desired level of confidence. If they want to keep the same level of confidence but divide the margin of error in third, how many people will they have to poll?

Answer:

$n^{-1/2} \propto ME$

$(1/3)ME = (1/3)n^{-1/2} = (1/9n)^{1/2}$
The EPA would need to poll 9 times as many people.

II. Significance Test for Proportions (7 x 3 = 21 pts)

1) Apple runs a quality control check on its products that come off the assembly line. Every 500th product is inspected for defects to make sure that the machines are calibrated correctly and working like they should. Due to machine deviations, they do expect 5% of products to be defective in a single day. The quality control specialist has to determine if a significant number of products is coming off the line defective.

(a) Define the parameter of interest and write the appropriate null and alternative hypotheses for the test that is described.

Answer:

p = proportion of all defective products that come off the assembly line
H_0: $p = 0.05$
H_a: $p > 0.05$

The quality control specialist takes an SRS of products and finds that the sample proportion of defective products is 0.058, which produces a P-value of 0.027.

(b) Interpret the P-value in the context of the problem.

Answer:

The p-value is the probability, assuming H_0 is true, that the test statistic (sample proportion of defective products) would take a value as large as or greater than actually observed. There is a probability of 0.027 of observing a proportion of 0.058 defective products, assuming the true proportion is 0.05.

(c) What conclusion would you draw at the $\alpha = 0.05$ level? At the $\alpha = 0.01$ level?

Answer:

At the $\alpha = 0.05$ level the result is not statistically significant.
At the $\alpha = 0.01$ level the result is statistically significant.

2) According to the CD website, 88% of US adults aged 65 – 74 are fully vaccinated against COVID-19 (as of 11/2021). A researcher suspects that this proportion is lower in their community. She selects an SRS of 125 adults aged 65 – 74 in her community and asks if they are fully vaccinated. 100 of them say that they are fully vaccinated. Does the researcher have convincing evidence that the proportion is lower in her community? Perform a significance test using a 5% level of significance. **Explain using the 4 step process**.

Answer:

Step 1.

State:
p_0 = proportion of US adults aged 65 – 74 that are fully vaccinated against COVID-19
p = proportion of adults aged 65 – 74 in the researcher's community that are fully vaccinated against COVID-19
H_0: $p = p_0$
H_a: $p < p_0$
$\alpha = 0.05$

Step 2.

Plan: if conditions are met, use a one-sample z test

Random: SRS stated
10%: assume 125 adults aged 65-74 is less than total in the researcher's community
Large counts: assuming H_0 is true, number of positives (110) and negatives (15) are both greater than 10

Step 3.

Do:
sample proportion p = 100/125 = 0.8
SE = sqrt(0.88*0.12/125) = 0.029
z = (0.8-0.88)/0.029 = -2.76
P(z < -2.76) = 0.00289
P-value = 0.003

Step 4.

Conclude:
Because the P-value of 0.003 is less than the significance value of 0.05, reject H_0. The researcher has convincing evidence that the proportion of adults aged 65-74 in her community is lower than the proportion of US adults that are fully vaccinated against COVID-19.

THE END